

# Fan-out - regex vs functional on sistema-sindico

Date: 2026-05-28 Models: 1 Tasks: 4 N values: 64, 200, 600 temperature: 0.7 use\_cache: False

Engine: kernel.subagent\_runtime.SubagentRuntime (PyPI simplicio-prompt 1.7.0). For each (model, task, N), N real parallel LLM calls through LaneWorkerPool; each output scored TWO ways: real PHPUnit (functional) and a regex pattern match (structural cheap proxy). The comparison shows where regex agrees with phpunit (cheap proxy works) and where regex inflates a pass that phpunit fails (regex is misleading).

## Per (model, N) headline

| Model                          | N   | fn per-att | rx per-att | fn modal | rx modal | Cost     |
|--------------------------------|-----|------------|------------|----------|----------|----------|
| Qwen/Qwen2.5-Coder-3B-Instruct | 64  | 32%        | 85%        | 2/4      | 4/4      | \$0.0000 |
| Qwen/Qwen2.5-Coder-3B-Instruct | 200 | 32%        | 83%        | 2/4      | 3/4      | \$0.0000 |
| Qwen/Qwen2.5-Coder-3B-Instruct | 600 | 51%        | 85%        | 2/3      | 3/3      | \$0.0000 |

## Per N (all models)

| N   | fn per-att | rx per-att | gap | fn modal | rx modal |
|-----|------------|------------|-----|----------|----------|
| 64  | 32%        | 85%        | +53 | 2/4      | 4/4      |
| 200 | 32%        | 83%        | +51 | 2/4      | 3/4      |
| 600 | 51%        | 85%        | +34 | 2/3      | 3/3      |

## Regex vs phpunit gap per task

| Task                    | fn  | rx  | gap (rx-fn) | verdict                         |
|-------------------------|-----|-----|-------------|---------------------------------|
| password_strength       | 74% | 99% | +25         | regex INFLATES (false positive) |
| password_require_symbol | 0%  | 61% | +61         | regex INFLATES (false positive) |
| env_get_int             | 55% | 96% | +41         | regex INFLATES (false positive) |
| env_get_bool            | 51% | 73% | +22         | regex INFLATES (false positive) |

## Interpretation

Functional (fn) = real phpunit exit code 0 on the FULL sindico suite including the per-case hidden test (or, for the bug-fix task, the existing PasswordPolicyTest). Regex (rx) = 'every structural pattern matched' on the generated file (per-task patterns in REGEX\_CHECKS\_BY\_TASK). Where rx >> fn the regex metric inflates results -- code looks right but doesn't run; where they agree, regex is a usable cheap proxy.